

National University of Computer and Emerging Sciences, Lahore Campus



Course: Electronic Devices & Circuits
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 11-Nov-16
 Section: ALL
 Exam: Midterm-2

Course Code: EE225
 Semester: Fall 2016
 Total Marks: 40
 Weight: 15%
 Page(s): 2
 Roll No:

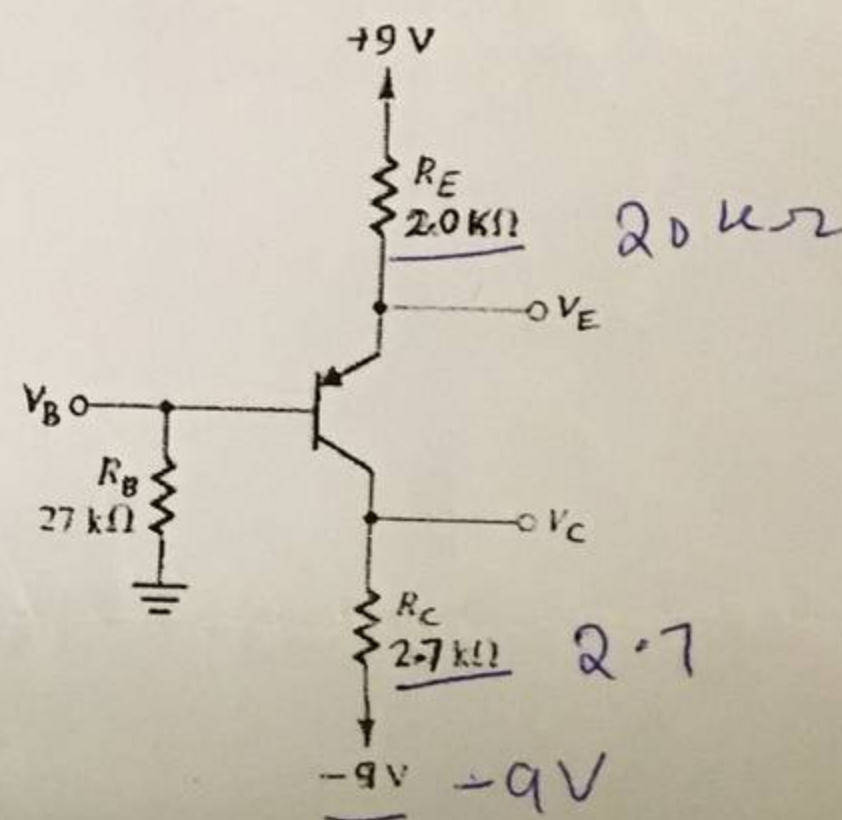
Instruction/Notes: Carefully read the paper and attempt each question (only once). Show all calculations, direct answers are not acceptable. Your answers should be approximated up to two decimal places. Attempt all parts of a question together.

Question # 1 [15 marks]

In the circuit shown, the transistor has

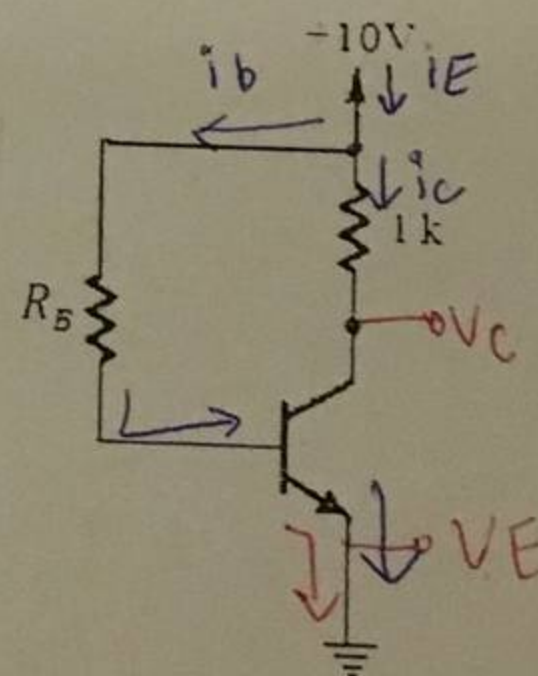
$$\beta = 30.$$

- Find the values of V_B , V_E and V_C ?
- Find the largest value to which R_C can be raised while the transistor remains in the active mode. - 21.07

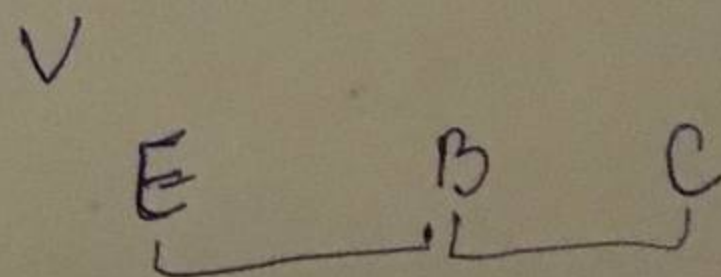


Question # 2 [10 marks]

For the circuit shown in figure, select a value of R_B so that the transistor saturates with an overdrive factor of 10. The BJT is specified to have a minimum β of 30. What is the value of forced β achieved? 3



$$30 \times 97.5$$



Question # 3 [15 marks]

A common emitter amplifier utilizes a BJT with $\beta = 200$ and $V_A = 200$ V. It is biased at $I_C = 0.5$ mA and has a collector resistance $R_C = 10$ k Ω .

- Draw and label the small signal equivalent circuit using Hybrid- π model including source and load resistance.
- Find R_{in} , R_o and A_{vo} ?
- If the amplifier is fed with a signal source having a resistance of 10 k Ω and a load resistance R_L of 10 k Ω is connected to the output terminal, Find A_v and G_v ?
- If a peak-to-peak output ac voltage of 1 V is measured at the collector, what ac input voltage and current must be associated with the base?